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UNIVERSITY OF
PLYMOUTH

MAL3018

COMPUTING PROJECT

Proposal Final Year Project

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Project Title

AI-Based HR Resume Recommender System

Keywords

HR, Resume, AI, NLP, Recommender System, Recruitment, Machine Learning, Resume Matching, Candidate Screening, Job Recommendation, Automation, Intelligent System

Link GitHub: <https://github.com/ImanHaqimi/hr-resume-recommender>

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Background and Problem Statement

The process of reviewing resumes manually is time-consuming, especially when HR departments receive hundreds of applications for each job opening. Many HR personnel struggle to efficiently identify candidates with the right qualifications and skills. Traditional keyword-based systems often fail to understand the true relevance between resumes and job requirements, which can lead to good candidates being overlooked.

The proposed **HR Resume Recommender System** aims to simplify this process using **Artificial Intelligence (AI)** and **Natural Language Processing (NLP)**. The system allows HR to post job vacancies and enables candidates to apply by submitting their resumes. The AI will analyze the resumes and automatically rank candidates based on their education level and skills that match the HR's job requirements. Resumes that do not meet the criteria will be filtered out, ensuring HR only reviews qualified applicants.

This project aims to make recruitment more efficient, accurate, and less prone to human bias, benefiting both HR and job seekers.

Objectives

- To design and develop a web-based system that allows HR to post job vacancies and candidates to submit resumes.
- To implement an AI model that analyzes and ranks candidates resumes based on the education level and skills matching the job description.
- To provide HR with a simple dashboard that displays ranked candidates and filters out those who do not meet the job requirements.

Scope of the Project

The HR Resume Recommender System will consist of two main roles: HR and Candidate.

In Scope:

- HR can post job vacancies and view ranked candidates.
- Candidates can apply for the jobs by uploading their resumes.
- The backend system will handle resume uploads, store data and communicate with the AI module
- AI will analyze education and skills, rank suitable candidates and automatically filter out unqualified ones.

Out of Scope:

- Resume builders or editing tools.
- Interview scheduling and chat features.

Literature Review / Related Works

Existing platforms like **LinkedIn** and **JobStreet** allow HR to collect resumes and perform basic keyword searches. While widely used, these systems often fail to capture the **contextual meaning** of qualifications, causing potentially suitable candidates to be overlooked.

Recent **AI and NLP advancements**, including **TF-IDF**, **Word2Vec**, and **BERT**, allow systems to understand text semantically, detect context, and identify relationships between candidate skills and job requirements.

The proposed system leverages these techniques to rank and filter candidates automatically. Compared to existing platforms, it provides:

System	Keyword-based?	Semantic Ranking?	Auto-filter?	Notes
LinkedIn	Yes	No	No	Standard job portal
JobStreet	Yes	No	No	Keyword search only
Proposed System	Yes + AI	Yes	Yes	Semantic AI ranking & filtering

This demonstrates that the proposed system improves HR efficiency, reduces bias and provides smarter candidates selection

Methodology

Development Approach: Agile (Scrum) with iterative testing and refinement.

Tools and Technologies:

- Frontend: HTML, CSS, JavaScript
- Backend: Node.js (Express)
- Database: MongoDB
- AI Module: Python (Flask/FastAPI), spaCy, TensorFlow, scikit-learn

System Flow:

1. HR posts a job vacancy.
2. Candidates upload resumes.
3. Backend stores resumes and job details.
4. AI module analyzes resumes:
 - Education Level (Degree or above)
 - Skill Match (keywords + semantic similarity)
5. Candidates are ranked; unqualified resumes filtered.
6. HR views ranked candidates on the dashboard.

Testing and Evaluation:

- Unit and integration testing.
- AI evaluation metrics: precision@k, recall, F1-score.
- User testing with sample HR scenarios.
- Candidate data anonymized during testing.

Expected Deliverables

- **Web Application:**
A fully functional HR Resume Recommender System accessible through a web browser, allowing HR to post job vacancies and candidates to submit resumes.
- **Backend System:**
Node.js and Express.js server with APIs connected to a MongoDB database for storing job postings, resumes, and candidate ranking data.
- **Frontend Interface:**
Responsive and user-friendly web interface for HR and candidates, featuring dashboards, resume submission forms, and ranked candidate views.
- **AI Module:**
Python-based AI module using NLP and machine learning (spaCy, TensorFlow, scikit-learn) to analyze resumes, rank candidates, and filter unqualified applicants.
- **Database Design:**
MongoDB database structure for storing job vacancies, resumes, candidate profiles, and ranking data.
- **Documentation:**
Complete project documentation including system design, architecture diagrams, AI workflow, methodology, and user guidance.
- **Testing Report:**
Results of unit, integration, and AI model evaluation, including ranking accuracy, precision metrics, and user testing feedback.
- **Final Presentation & Report:**
A summarized project report and presentation slides demonstrating the system, methodology, and results for evaluation.

Risk Plan

The risk plan identifies potential challenges that could impact the success of the project. Each risk is assessed based on Likelihood (how probable it is to occur) and Impact (the severity of its consequences). These two factors determine the Risk Level (High, Medium, Low).

- High Risk: Requires immediate attention and proactive mitigation strategies.
- Medium Risk: Needs monitoring and preventive measures to avoid escalation.
- Low Risk: Minimal impact but should still be tracked to prevent surprises.

The Mitigation Strategy column outlines actions to reduce the likelihood or impact of each risk. For example:

- Technical risks like Low AI ranking accuracy are addressed by starting with a baseline model and iterative testing.
- Operational risks such as Schedule slippage are managed through Agile sprints and progress tracking.
- Ethical risks like Candidate data privacy are mitigated by anonymizing data and following compliance standards.

The accompanying Risk Matrix diagram visually represents the relationship between likelihood and impact, helping prioritize which risks need the most attention.

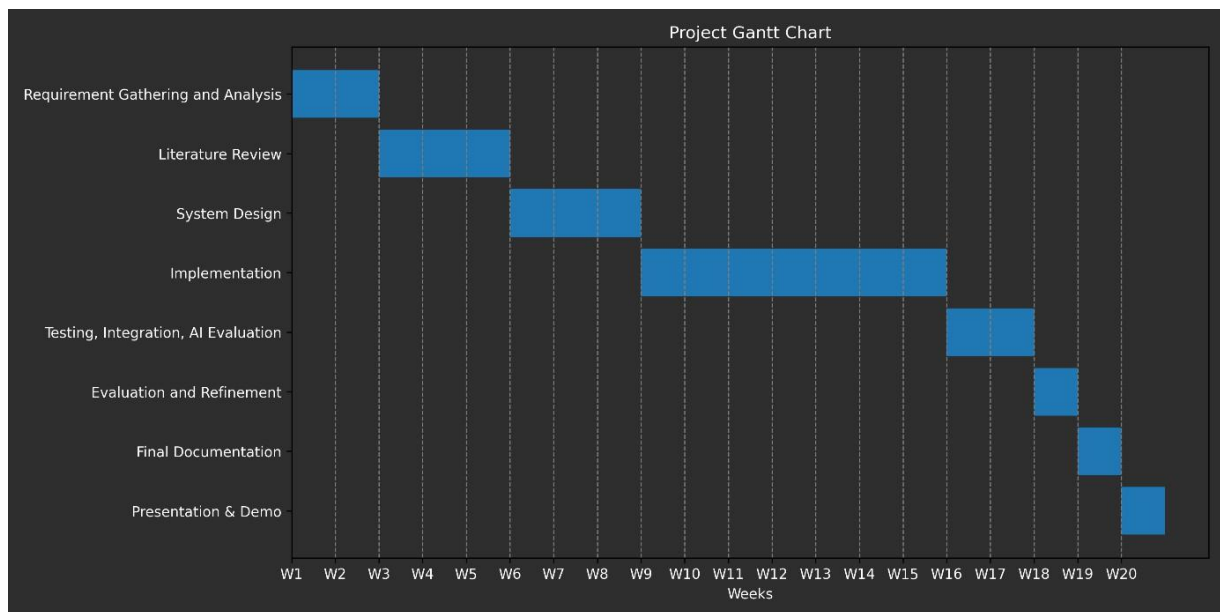
Risk Plan

#	POTENTIAL RISK	LIKEHOOD	IMPACT	MITIGATION STRATEGY
1	Low AI ranking accuracy	Medium	High	Start with a simple baseline model (e.g., TF-IDF + Cosine Similarity). Iteratively test and improve using precision@k, recall, F1-score.
2	Integration issues (Node.js & Python AI module)	Medium	Medium	Define clear API endpoints, test integration early and frequently, use mock data for initial testing.
3	Insufficient dataset for AI training	Medium	High	Use publicly available datasets (e.g., Kaggle), data augmentation, document limitations in the report.
4	Scope creep	High	Medium	Strictly follow in-scope/out-of-scope items, get supervisor approval before adding features.
5	Schedule slippage	Medium	High	Use Agile sprints, prioritize core features, track progress weekly.
6	Hardware failure / data loss	Low	High	Backup all code and data to GitHub and cloud storage regularly.
7	Candidate data privacy / ethical concerns	Medium	High	Anonymize candidate data, follow ethical standards, ensure no sensitive info is exposed.
8	Low user adoption / HR resistance	Medium	Medium	Conduct early user testing with HR, gather feedback, provide training/demo sessions.

Risk Matrix

Likelihood	High	Low	High	High
	Medium	Moderate	Moderate	Moderate
	Low	Low	Moderate	Moderate
		Low	Medium	High
		Impact		

Proposed Gantt Chart



Gantt Chart Explanation

The visual Gantt chart above illustrates the project timeline, showing tasks and their durations across 20 weeks. Each phase is arranged sequentially, starting from requirement gathering and ending with the final presentation. This layout provides a clear overview of the workflow and helps track progress effectively.

The chart highlights:

- **Duration of each task** in weeks
- **Order of phases** for systematic development
- **Key milestones** for monitoring progress

By following this timeline, the project ensures clarity, efficiency, and flexibility for iterative improvements during testing and evaluation.

Week(s)	Task / Milestone	Expected Output
1-2	Requirement Gathering and Analysis	Requirement Specification Document
3-5	Literature Review	Literature Review Chapter
6-8	System Design	System Design Document, ERD, UI Mockups
9-15	Implementation	Functional System Components
16-17	Testing, Integration, AI Evaluation	Test Reports, Performance Metrics
18	Evaluation and Refinement	System Refinement
19	Final Documentation	Final Report
20	Presentation C Demo	Slides, Live Demo

Resources Required

Hardware:

- Personal laptop or desktop with at least **Intel i5 / Ryzen 5** processor.
- Minimum **8GB RAM** (16GB recommended for AI processing).
- **256GB SSD** storage or higher.
- Reliable **internet connection** for API access and cloud backups.

Software:

- **Frontend:** HTML, CSS, JavaScript.
- **Backend:** Node.js, Express.js.
- **Database:** MongoDB (local or cloud).
- **Server Environment:** Local Node.js setup for development and testing.
- **Code Editing:** Visual Studio Code.
- **Testing Tools:** Postman (API testing), Jest or Mocha (unit testing).
- **Version Control:** Git and GitHub.
- **Design Tools:** Figma / Canva for mockups, diagrams, and dashboards.

Datasets / APIs:

- Public datasets for sample resumes and job descriptions (Kaggle, academic sources).
- Sample JSON or CSV files for AI ranking and filtering tests.

Additional Tools:

- Documentation tools: Google Docs / Microsoft Word.
- Presentation tools: Microsoft PowerPoint / Canva.

Expected Outcome and Significance

The HR Resume Recommender System will provide HR departments with an **intelligent, automated tool** that ranks and filters job applicants based on their **education, skills, and job requirement relevance**. This outcome will lead to several key benefits:

1. **Efficiency Improvement:** HR staff will save time by only reviewing **pre-selected, highly relevant candidates**, reducing the hours spent on manual resume screening.
2. **Improved Accuracy:** By leveraging AI and NLP, the system can **identify candidate qualifications and skills more precisely**, reducing the risk of overlooking suitable applicants due to keyword limitations or human error.
3. **Reduction of Human Bias:** The AI-driven ranking helps **minimize unconscious bias** in candidate selection, promoting fairer recruitment practices.
4. **Enhanced Candidate Experience:** Applicants receive **faster consideration** for jobs, as resumes are processed and matched more quickly, improving communication and transparency.
5. **Practical Application of AI in HR:** The project demonstrates **real-world implementation of AI and NLP techniques**, bridging the gap between research and practice. HR professionals can gain insights into how AI can support recruitment decision-making.
6. **Academic Contribution:** The system contributes to knowledge in **AI-driven recruitment, NLP resume analysis, and recommender systems**, providing a framework for future research and potential improvements in the automation of HR processes.
7. **Scalability and Future Extensions:** The modular design allows the system to scale to **multiple job postings and high applicant volumes**, and can be extended with additional features, such as **skill gap analysis, predictive hiring success, or integration with interview scheduling**.

References

- LinkedIn Talent Solutions. (2025) *The Future of Recruiting 2025*. Available at: <https://business.linkedin.com/talent-solutions/resources/future-of-recruiting>
- JobStreet Malaysia. (2025) *Understanding the recruitment process: strategies for success*. Available at: <https://my.jobstreet.com/career-advice/article/recruitment-process-strategies-for-success>
- MongoDB. (2025) *MERN Stack Explained*. Available at: <https://www.mongodb.com/resources/languages/mern-stack> .
- Kaggle. (2024) *Resume Dataset for AI/ML Applications*. Available at: <https://www.kaggle.com/datasets/resumes>